

CS7.401 Introduction to NLP Quiz 1 (Spring 2026) Answer Key - All Sets

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1 Gradient Calculation Question No 4 or 5

Loss function $\mathbf{x}$ is $\mathcal{L}(\Theta) = \theta_1^2 + \theta_2\theta_3$

- 1 mark

$$\boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t - \eta \nabla L(\boldsymbol{\theta}_t) \quad (1)$$

$$\theta_1^{t+1} = \theta_1^t - \eta \frac{\partial L}{\partial \theta_1}; \theta_2^{t+1} = \theta_2^t - \eta \frac{\partial L}{\partial \theta_2}; \theta_3^{t+1} = \theta_3^t - \eta \frac{\partial L}{\partial \theta_3} \quad (2)$$

$$\theta_1^{t+1} = \theta_1^t - 2\eta\theta_1^t; \theta_2^{t+1} = \theta_2^t - \eta\theta_3^t; \theta_3^{t+1} = \theta_3^t - \eta\theta_2^t \quad (3)$$

- 6 marks - 1 mark each for each gradient calculation - total 3 marks for first gradient and 3 for second gradient
- What value do they eventually converge to? - 1 mark
- Loss neither minimizes or maximizes locally but reaches saddle point - 2 marks

2 BPE Question No 4 or 5

- Each iteration - 2.5 marks
- (Pair to merge = .25 marks; Frequency = .25 marks; Updated vocab = 1 mark; Updated corpus = 1 mark) - per merge
- BPE typically breaks ties left-to-right but we are allowing picking any option
- Also allowing for -, _, eos etc. in vocab as long as all merges are consistent

Sample answer for Telugu

Iteration 1

1. **Pair to merge:** (n, a)
 2. **Frequency:** 50
(*Calculation: $2 \times 12 + 2 \times 6 + 1 \times 10 + 1 \times 4 = 50$*)
 3. **New Token & Vocabulary:** The new token is na.
Updated Vocabulary: { n, a, d, u, s, t, i, r, -, _, na }
 4. **Corpus after the merge:**
 - na d u - s t u n n na d u _ (Freq: 12)
 - na d u - s t u n - na r u _ (Freq: 6)
 - t i n t u - n na d u _ (Freq: 10)
 - t i n t u - n na r u _ (Freq: 4)
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Iteration 2

1. **Pair to merge:** (na, d)
(*Note: (na, d) and (d, u) both have a frequency of 40.*)
 2. **Frequency:** 40
(*Calculation: $2 \times 12 + 1 \times 6 + 1 \times 10 + 0 \times 4 = 40$*)
 3. **New Token & Vocabulary:** The new token is nad.
Updated Vocabulary: { n, a, d, u, s, t, i, r, -, _, na, nad }
 4. **Corpus after the merge:**
 - nad u - s t u n n nad u _ (Freq: 12)
 - nad u - s t u n - na r u _ (Freq: 6)
 - t i n t u - n nad u _ (Freq: 10)
 - t i n t u - n na r u _ (Freq: 4)
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Iteration 3

1. **Pair to merge:** (nad, u)
2. **Frequency:** 40
(*Calculation: $2 \times 12 + 1 \times 6 + 1 \times 10 + 0 \times 4 = 40$*)

3. **New Token & Vocabulary:** The new token is nadu.
Updated Vocabulary: { n, a, d, u, s, t, i, r, -, _, na, nad, nadu }
 4. **Corpus after the merge:**
 - nadu - s t u n n nadu _ (Freq: 12)
 - nadu - s t u n - na r u _ (Freq: 6)
 - t i n t u - n nadu _ (Freq: 10)
 - t i n t u - n na r u _ (Freq: 4)
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Iteration 4

1. **Pair to merge:** (t, u)
2. **Frequency:** 32
(Calculation: $1 \times 12 + 1 \times 6 + 1 \times 10 + 1 \times 4 = 32$)
3. **New Token & Vocabulary:** The new token is tu.
Updated Vocabulary: { n, a, d, u, s, t, i, r, -, _, na, nad, nadu, tu }
4. **Corpus after the merge:**
 - nadu - s tu n n nadu _ (Freq: 12)
 - nadu - s tu n - na r u _ (Freq: 6)
 - t i n tu - n nadu _ (Freq: 10)
 - t i n tu - n na r u _ (Freq: 4)